~ Calibration Certificate ~

Model Number: 480B21

Serial Number: 41481

Description: 3 Channel Signal Conditioner

Manufacturer: PCB

Customer:

P.O.:

Method : Comparison Method (AT103-19)

Calibration Data

Temperature: 75 °F (24 °C)

Humidity: 43%

Channel	Volts	Current (mA)	Gain X1	Gain X10	Gain X100
1	27.3	2.97	1.001	10.011	100.1
2	26.4	2.98	1.001	10.008	100.1
3	26.4	2.97	1.001	10.013	100.1

Condition of Unit

(Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.)

As Found: N/A

As Left: New unit, in tolerance

Notes

1. Calibration is N.I.S.T. traceable through PCB control number QC726.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement uncertainty (95% confidence level with a coverage factor of 2) for DC Volts and AC Volts/Gain is +/- 0.13% of reading.
5. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.

Technician: Hannah Lynch HCL 5512

Date: 02/17/25

Due Date:

ilac-MRA ACCREDITED

CALIBRATION CERT #1862.02

PCB PIEZOTRONICS
AN AMPHENOL COMPANY

Headquarters: 3425 Walden Avenue, Depew, NY 14043

Calibration Performed at: 10869 Highway 903, Halifax, NC 27839

TEL: 888-684-0013 FAX: 716-685-3886 www.pcb.com

ELE98-3822700065

~ Calibration Certificate ~

Per ISO 16063-21

Model Number: **352C22**

Serial Number: **LW521801**

Description: ICP® Accelerometer

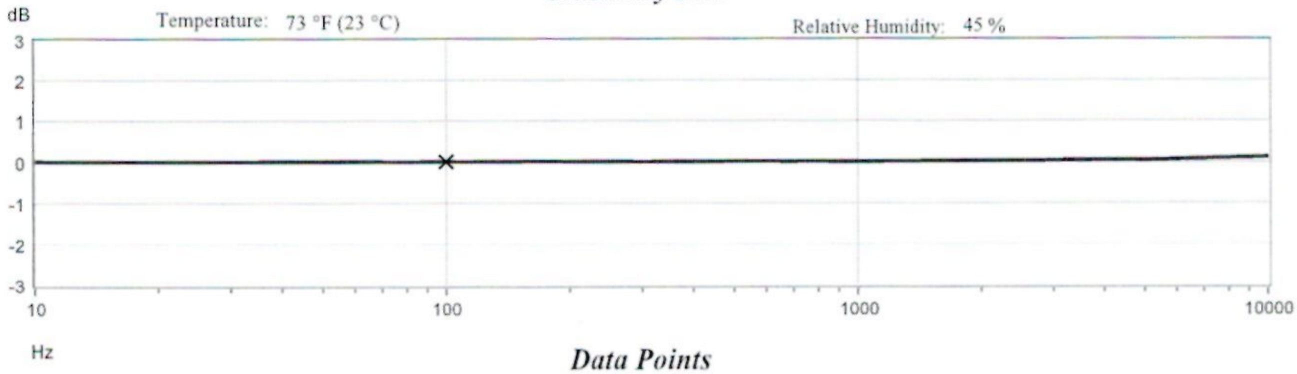
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz	**10.56 mV/g**	Output Bias	10.5 VDC
	(1.077 mV/m/s²)	Transverse Sensitivity	2.1 %
Discharge Time Constant	2.1 seconds	Resonant Frequency	82.5 kHz

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	0.0	300	-0.0	7000	0.7
15	-0.1	500	0.1	10000	1.2
30	-0.0	1000	0.0		
50	0.0	3000	0.3		
REF. FREQ.	0.0	5000	0.4		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical
Acceleration Level (pk)[1]: 10.0 g (98.1 m/s²)
[1]The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².
[2]The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

1. Calibration is NIST Traceable thru Project 684/O-0000035821 and PTB Traceable thru Project 17016.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows: 5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician: Alexander Stanley APS 5807 Date: 2/27/2025

ilac-MRA ACCREDITED CALIBRATION CERT#: 1862.02

PCB PIEZOTRONICS
AN AMPHENOL COMPANY

HEADQUARTERS: 3425 WALDEN AVENUE - DEPEW, NY 14043
CALIBRATION PERFORMED AT: 10869 HIGHWAY 903 - HALIFAX, NC 27839
TEL: +1 (888) 684-0013 - FAX: +1 (716) 685-3886 - www.pcb.com

CAL48-3823557793.510+0

ACS-1

~ *Certificate of Calibration and Compliance* ~

Model :	378B02	**Manufacturer :**	PCB
Serial :	173314	**Description :**	1/2" Free-Field Microphone and Preamplifier

Microphone Model :	377B02	**Serial :**	364175	**Manufacturer :**	PCB
Preamplifier Model :	426E01	**Serial :**	090621	**Manufacturer :**	PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0146	CA2115	07/11/2024	07/11/2025
Larson Davis	PRM902	4701	CA1450	07/11/2024	07/11/2025
Larson Davis	PRM916	0129	CA1084	08/16/2024	08/15/2025
Larson Davis	CAL250	4147	LD018	07/08/2024	07/08/2025
Larson Davis	2201	151	CA2073	09/05/2024	09/05/2025
Larson Davis	GPRM902	5337	CA2153	10/25/2024	10/24/2025
Larson Davis	PRM915	132	CA1552	09/11/2024	09/11/2025
Larson Davis	PRA951-4	0241	CA1449	08/17/2024	08/15/2025
Bruel & Kjaer	4192	3259548	CA3533	10/11/2024	10/10/2025
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.20 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik Date: 03/05/2025

ilac-MRA ACCREDITED CALIBRATION CERT#: 1862.01

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ID: CAL112-3824017102.262+0

~ *Calibration Report* ~

Model : 378B02 **Manufacturer :** PCB

Serial : 173314 **Description :** 1/2" Free-Field Microphone and Preamplifier

Microphone Model : 377B02 **Serial :** 364175 **Manufacturer :** PCB

Preamplifier Model : 426E01 **Serial :** 090621 **Manufacturer :** PCB

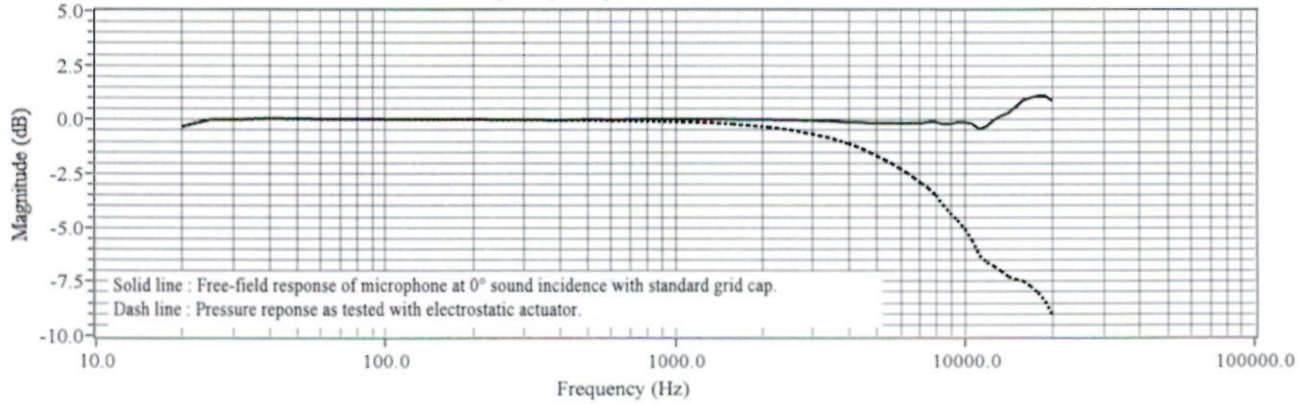
Calibration Data

System Sensitivity at 251.2 Hz : 53.67 mV/Pa
-25.41 dB re 1 V/Pa

Polarization Voltage, External : 0 V

Temperature: 68 °F (20 °C) Ambient Pressure: 976 mbar Relative Humidity: 36 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)
20.00	-0.31	-0.31	1584.90	-0.20	0.01	6683.40	-2.69	-0.17
25.10	0.02	0.02	1678.80	-0.23	-0.00	7079.50	-2.94	-0.16
31.60	0.02	0.02	1778.30	-0.25	0.00	7498.90	-3.18	-0.11
39.80	0.06	0.06	1883.60	-0.28	-0.00	7943.30	-3.49	-0.10
50.10	0.04	0.04	1995.30	-0.31	-0.00	8414.00	-3.93	-0.20
63.10	0.03	0.03	2113.50	-0.34	0.00	8912.50	-4.31	-0.20
79.40	0.02	0.02	2238.70	-0.38	-0.01	9440.60	-4.64	-0.12
100.00	0.02	0.02	2371.40	-0.42	-0.01	10000.00	-5.07	-0.12
125.90	0.01	0.01	2511.90	-0.48	-0.02	10592.50	-5.58	-0.18
158.50	0.01	0.01	2660.70	-0.52	-0.01	11220.20	-6.29	-0.43
199.50	0.01	0.01	2818.40	-0.60	-0.04	11885.00	-6.62	-0.30
251.20	0.00	0.00	2985.40	-0.65	-0.03	12589.30	-6.79	-0.02
316.20	-0.00	0.01	3162.30	-0.74	-0.06	13335.20	-7.01	0.18
398.10	-0.01	-0.01	3349.70	-0.81	-0.07	14125.40	-7.28	0.31
501.20	-0.02	0.02	3548.10	-0.89	-0.07	14962.40	-7.40	0.57
631.00	-0.04	0.00	3758.40	-1.00	-0.10	15848.90	-7.47	0.88
794.30	-0.06	0.03	3981.10	-1.11	-0.11	16788.00	-7.71	1.01
1000.00	-0.09	0.03	4217.00	-1.24	-0.13	17782.80	-8.02	1.09
1059.30	-0.10	0.03	4466.80	-1.37	-0.14	18836.50	-8.42	1.09
1122.00	-0.10	0.04	4731.50	-1.53	-0.16	19952.60	-9.07	0.86
1188.50	-0.12	0.03	5011.90	-1.69	-0.16			
1258.90	-0.13	0.03	5308.80	-1.86	-0.16			
1333.50	-0.14	0.04	5623.40	-2.04	-0.16			
1412.50	-0.16	0.03	5956.60	-2.25	-0.18			
1496.20	-0.18	0.02	6309.60	-2.46	-0.17			

Technician: Leonard Lukasik Date: 03/05/2025

ilac-MRA
A2LA ACCREDITED CALIBRATION CERT#: 1862.01

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ID: CAL112-3824017102.262+0

~ Calibration Certificate ~

Model Number: 086E80

Serial Number: 52563

Description: Impulse Force Hammer

Manufacturer: PCB

Customer:

P.O. Number:

Method: Impulse (AT303-1)

Calibration Data

Output Bias: 10.5 VDC

Temperature: 73 °F = 23 °C

Relative Humidity: 39 %

Hammer Configuration		***Hammer Sensitivity****	
Tip	**Extender**	**mV/lb**	**mV/N**
Steel	None	110.9	24.94
Vinyl	Steel	91.71	20.62
Vinyl	None	85.44	19.21

*Above data is valid for all supplied tips.

Condition of Unit

As Found: N/A

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Notes:

1. Calibration is N.I.S.T. Traceable through Project 684/O-0000069046 and PTB Traceable through Project 17036.
2. This certificate may not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with a coverage factor of 2) is +/-3.8%.

Technician: Mary Warren MW

Date: 3/4/2025

ilac-MRA ACCREDITED CALIBRATION CERT# 1862.01

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CAL13-3823956603.413+0

~ Calibration Certificate ~

Per ISO 16063-21

Model Number: 352A73

Serial Number: LW521679

Description: ICP® Accelerometer

Manufacturer: PCB

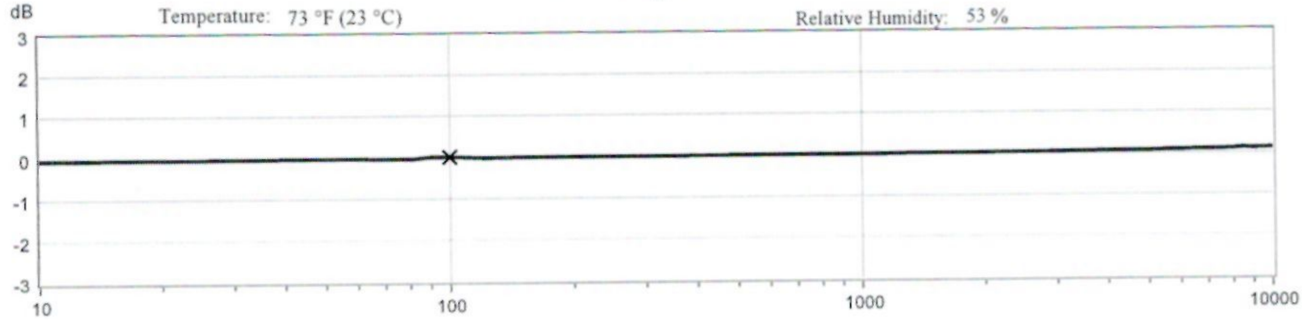
Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz	**5.58 mV/g**	Output Bias	9.3 VDC
	(0.569 mV/m/s²)	Transverse Sensitivity	0.7 %
Discharge Time Constant	0.5 seconds	Resonant Frequency	108.1 kHz

Sensitivity Plot

Temperature: 73 °F (23 °C) Relative Humidity: 53 %



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-0.7	300	-0.2	7000	0.9
15	-0.6	500	-0.0	10000	1.2
30	-0.5	1000	0.0		
50	-0.4	3000	0.3		
REF. FREQ.	0.0	5000	0.6		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical

Acceleration Level (pk)[1]: 10.0 g (98.1 m/s²)[2]

[1]The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².

[2]The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

1. Calibration is NIST Traceable thru Project 684/O-0000035821 and PTB Traceable thru Project 17016.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows: 5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician: Jaquan Robinson JR Date: 3/10/2025

ilac-MRA A2LA ACCREDITED CALIBRATION CERT#: 1862.02

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CAL48-3824504149.730+0

ACS-1

~ *Certificate of Calibration and Compliance* ~

Model :	378C10	**Manufacturer :**	PCB
Serial :	175491	**Description :**	1/4" Pressure Microphone and Preamplifier

Microphone Model :	377C10	**Serial :**	363727	**Manufacturer :**	PCB
Preamplifier Model :	426B03	**Serial :**	082728	**Manufacturer :**	PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	2699	TA468	12/27/2024	12/26/2025
Larson Davis	PRM916	0129	CA1084	08/16/2024	08/15/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	151	CA2073	09/05/2024	09/05/2025
Larson Davis	GPRM902	5337	CA2153	10/25/2024	10/24/2025
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	08/17/2024	08/15/2025
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik Date: 07/14/2025

ilac-MRA ACCREDITED CALIBRATION CERT# 1862.01

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3425 WALDEN AVENUE - DEPEW, NY 14043
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ID: CAL112-3835349575.439+0

~ *Calibration Report* ~

Model : 378C10 **Manufacturer :** PCB

Serial : 175491 **Description :** 1/4" Pressure Microphone and Preamplifier

Microphone Model : 377C10 **Serial :** 363727 **Manufacturer :** PCB

Preamplifier Model : 426B03 **Serial :** 082728 **Manufacturer :** PCB

Calibration Data

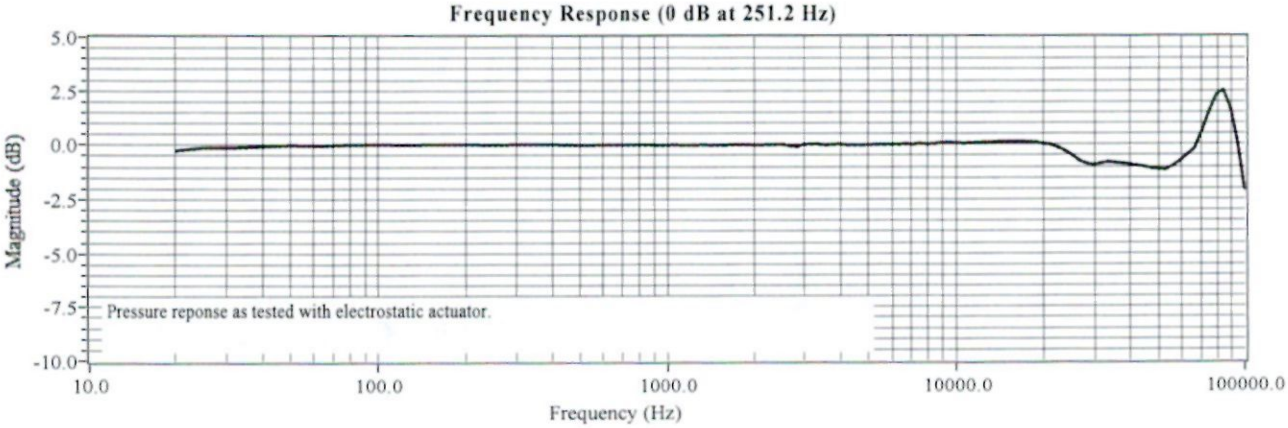
System Sensitivity at 251.2 Hz : 0.97 mV/Pa

-60.27 dB re 1 V/Pa

Polarization Voltage, External : 0 V

Temperature: 70 °F (21 °C) Ambient Pressure: 991 mbar Relative Humidity: 49 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.25	1584.90	-0.02	6683.40	0.03	28250.00	-0.88
25.10	-0.12	1678.80	-0.00	7079.50	0.02	29750.00	-0.93
31.60	-0.12	1778.30	-0.01	7498.90	0.05	31500.00	-0.88
39.80	-0.07	1883.60	0.00	7943.30	0.03	33500.00	-0.81
50.10	-0.02	1995.30	-0.02	8414.00	0.05	35500.00	-0.85
63.10	-0.04	2113.50	-0.01	8912.50	0.09	37500.00	-0.88
79.40	0.00	2238.70	-0.01	9440.60	0.10	39750.00	-0.92
100.00	0.01	2371.40	0.00	10000.00	0.09	42250.00	-0.98
125.90	0.00	2511.90	0.01	10592.50	0.06	44750.00	-1.02
158.50	0.01	2660.70	-0.04	11220.20	0.07	47250.00	-1.10
199.50	0.01	2818.40	-0.09	11885.00	0.10	50000.00	-1.13
251.20	0.00	2985.40	0.02	12589.30	0.10	53000.00	-1.15
316.20	0.02	3162.30	0.04	13335.20	0.10	56250.00	-0.98
398.10	0.01	3349.70	0.02	14125.40	0.13	59500.00	-0.76
501.20	-0.02	3548.10	-0.02	14962.40	0.14	63000.00	-0.47
631.00	-0.01	3758.40	0.01	15848.90	0.13	66750.00	-0.20
794.30	-0.01	3981.10	0.03	16788.00	0.13	70750.00	0.60
1000.00	-0.02	4217.00	-0.01	17782.80	0.12	75000.00	1.54
1059.30	-0.01	4466.80	-0.02	18836.50	0.10	79500.00	2.31
1122.00	-0.01	4731.50	-0.01	19952.60	0.04	84250.00	2.52
1188.50	-0.01	5011.90	0.00	21250.00	-0.00	89250.00	1.66
1258.90	-0.02	5308.80	0.01	22500.00	-0.14	94500.00	0.08
1333.50	-0.01	5623.40	0.01	23750.00	-0.28	100000.00	-2.09
1412.50	-0.02	5956.60	0.02	25000.00	-0.47		
1496.20	-0.02	6309.60	0.01	26500.00	-0.72		

Technician: Leonard Lukasik Date: 07/14/2025

ilac-MRA ACCREDITED CALIBRATION CERT#: 1862.01

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ID: CAL112-3835349575.439+0

~ *Certificate of Calibration and Compliance* ~

Model :	378C10	**Manufacturer :**	PCB
Serial :	175492	**Description :**	1/4" Pressure Microphone and Preamplifier

Microphone Model :	377C10	**Serial :**	363725	**Manufacturer :**	PCB
Preamplifier Model :	426B03	**Serial :**	082729	**Manufacturer :**	PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	2699	TA468	12/27/2024	12/26/2025
Larson Davis	PRM916	0129	CA1084	08/16/2024	08/15/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	151	CA2073	09/05/2024	09/05/2025
Larson Davis	GPRM902	5337	CA2153	10/25/2024	10/24/2025
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	08/17/2024	08/15/2025
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik Date: 07/14/2025

ilac-MRA

ACCREDITED CALIBRATION CERT# 1862.01

PCB PIEZOTRONICS
AN AMPHENOL COMPANY
3425 WALDEN AVENUE - DEPEW, NY 14043
TEL: +1 (888) 684-0013 - FAX: +1 (716) 685-3886 - www.pcb.com

ID: CAL112-3835348773.530+0

~ *Calibration Report* ~

Model : 378C10 **Manufacturer :** PCB

Serial : 175492 **Description :** 1/4" Pressure Microphone and Preamplifier

Microphone Model : 377C10 **Serial :** 363725 **Manufacturer :** PCB

Preamplifier Model : 426B03 **Serial :** 082729 **Manufacturer :** PCB

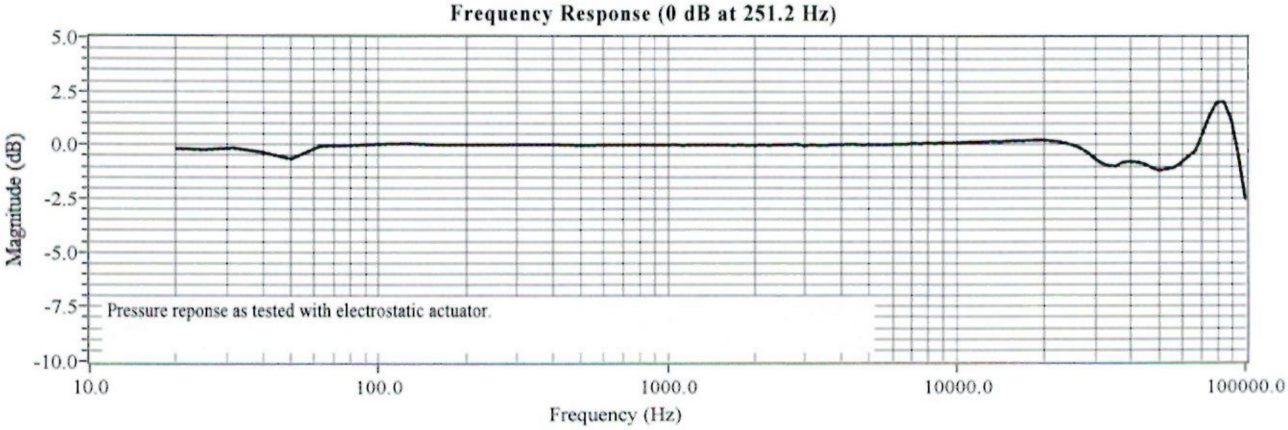
Calibration Data

System Sensitivity at 251.2 Hz : 1.01 mV/Pa
-59.92 dB re 1 V/Pa

Polarization Voltage, External : 0 V

Temperature: 70 °F (21 °C) Ambient Pressure: 991 mbar Relative Humidity: 49 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.17	1584.90	0.00	6683.40	0.03	28250.00	-0.34
25.10	-0.21	1678.80	-0.01	7079.50	0.07	29750.00	-0.59
31.60	-0.15	1778.30	0.00	7498.90	0.05	31500.00	-0.84
39.80	-0.35	1883.60	-0.01	7943.30	0.09	33500.00	-0.97
50.10	-0.66	1995.30	-0.02	8414.00	0.07	35500.00	-1.00
63.10	-0.06	2113.50	-0.01	8912.50	0.10	37500.00	-0.82
79.40	-0.03	2238.70	-0.02	9440.60	0.10	39750.00	-0.78
100.00	0.03	2371.40	0.00	10000.00	0.10	42250.00	-0.81
125.90	0.05	2511.90	-0.01	10592.50	0.12	44750.00	-0.90
158.50	0.01	2660.70	0.02	11220.20	0.13	47250.00	-1.08
199.50	0.01	2818.40	0.04	11885.00	0.13	50000.00	-1.19
251.20	0.00	2985.40	-0.03	12589.30	0.15	53000.00	-1.14
316.20	0.02	3162.30	-0.01	13335.20	0.16	56250.00	-1.06
398.10	0.01	3349.70	-0.02	14125.40	0.15	59500.00	-0.86
501.20	-0.02	3548.10	-0.03	14962.40	0.18	63000.00	-0.56
631.00	0.01	3758.40	0.01	15848.90	0.19	66750.00	-0.30
794.30	-0.00	3981.10	0.01	16788.00	0.19	70750.00	0.51
1000.00	-0.00	4217.00	0.04	17782.80	0.21	75000.00	1.38
1059.30	0.01	4466.80	0.03	18836.50	0.23	79500.00	1.99
1122.00	-0.01	4731.50	0.00	19952.60	0.22	84250.00	2.01
1188.50	0.01	5011.90	0.01	21250.00	0.17	89250.00	1.15
1258.90	-0.01	5308.80	0.02	22500.00	0.15	94500.00	-0.47
1333.50	0.01	5623.40	0.02	23750.00	0.09	100000.00	-2.55
1412.50	-0.01	5956.60	0.03	25000.00	0.00		
1496.20	0.01	6309.60	0.03	26500.00	-0.12		

Technician: Leonard Lukasik Date: 07/14/2025

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CALIBRATION CERT#: 1862.01

PCB PIEZOTRONICS
AN AMPHENOL COMPANY
3425 WALDEN AVENUE - DEPEW, NY 14043
TEL: +1 (888) 684-0013 - FAX: +1 (716) 685-3886 - www.pcb.com

ID: CAL112-3835348773.530+0

~ Calibration Certificate ~

Model Number: 086C01

Serial Number: 52635

Description: Impulse Force Hammer

Manufacturer: PCB

Customer:

P.O. Number:

Method: Impulse (AT303-1)

Calibration Data

Output Bias: 10.2 VDC

Temperature: 73 °F = 23 °C

Relative Humidity: 43 %

Hammer Configuration		*Hammer Sensitivity**	
Tip	**Extender**	**mV/lb**	**mV/N**
Plastic/Vinyl with tuning mass	None	51.65	11.61
Plastic/Vinyl with tuning mass	Aluminum	53.29	11.98
Plastic/Vinyl	None	47.46	10.67

*Above data is valid for all supplied tips.

Condition of Unit

As Found: N/A

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Notes:

1. Calibration is N.I.S.T. Traceable through Project 684/O-0000069046 and PTB Traceable through Project 17036.
2. This certificate may not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with a coverage factor of 2) is +/-3.8%.

Technician: Austin Crawley AC

Date: 2/26/2025

ilac-MRA

ACCREDITED
CALIBRATION CERT#: 1862.02

PCB PIEZOTRONICS
AN AMPHENOL COMPANY

HEADQUARTERS: 3425 WALDEN AVENUE - DEPEW, NY 14043
CALIBRATION PERFORMED AT: 10869 HIGHWAY 903 - HALIFAX, NC 27839
TEL: +1 (888) 684-0013 - FAX: +1 (716) 685-3886 - www.pcb.com

CAL58-3823444942.009+0

Calibration Certificate

Polytec

Certificate No.: C131563 / L1030877

Calibration Object:	Laser Vibrometer
Manufacturer:	Polytec, GmbH
Controller/Front-End Type:	VFX-F-110
Controller/Front End SN:	0331036
Sensor Head Type:	VFX-I-160
Sensor Head SN:	0331024
Calibration Procedure:	Vib_Calibration_1217_02
Date of Calibration:	July 21, 2025
Calibration Due Date: *Confirmed by customer?*	July 21, 2027 *Yes*
Technician:	EMh
Calibration Location:	Polytec, Inc
Customer:	**Massachusetts Institute of Technology** 350 Brookline St Middlesex Cambridge, MA 02139 United States

λ=1550 nm

Polytec, Inc
1 Cabot Road, Suite 102
Hudson, MA 01749
United States
Telephone: 508-417-1040
Fax: 508-281-4725

Traceability:
The calibrations within the certificate/report are traceable through NIST or another National Metrology Institute to the International System of Units (SI).

This calibration certificate includes the following protocols:

Page	Object	Procedure
2	VFX-V (X)	Part 1: Electrical Calibration
3	VFX-V (X)	Part 2: Mechanical End Test
4	VFX-D (X)	Part 1: Electrical Calibration
5	VFX-D (X)	Part 2: Mechanical End Test
6	VFX-I-160	Alignment of Head Protocol

Note: The statement of compliance in this certificate was issued without taking the uncertainty of measurement into consideration. The customer shall assess the results and uncertainty when determining if the results meet their needs.

Sign:

Elizabeth Mhaurwala

EMh

Velocity Decoder Calibration Protocol

Part 1: Electrical Calibration

Polytec

Certificate No.: C131563 / L1030877

Device Type: VFX-F-110 **Serial No.:** 0331036

Decoder Type: VFX-V (X)

Reference Conditions:

Frequency[1]:	1 kHz
Amplitude[1]:	70%
Target Temp.:	(25 ± 3)°C
Actual Temp:	23.1 °C
Uncertainty[2]:	1.00%

Measurement Procedure:
Synthetic Doppler signals substitute those which would come from the vibrometer optical system when acquiring a sinusoidal vibration of known frequency and amplitude. The output of the respective decoder is compared to the represented reference vibration amplitude.

[1]In some measurement ranges, deviating values are possible (refer to table)
[2]The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Tektronix AFG 3252	Type:	Keithley 2001
Serial Number:	C022189	Serial Number:	0661057
Certificate No.:	3292055	Certificate No.:	3287284
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
10.0 mm/s	7.0000 mm/s @ 1 kHz	0.9900 Vrms	0.9886 Vrms
20.0 mm/s	14.000 mm/s @ 1 kHz	0.9900 Vrms	0.9886 Vrms
50.0 mm/s	35.000 mm/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
100 mm/s	70.000 mm/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
200 mm/s	140.00 mm/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms
500 mm/s	350.00 mm/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
1.00 m/s	700.00 mm/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms
2.00 m/s	1.4000 m/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms
5.00 m/s	3.5000 m/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
6.00 m/s	4.2000 m/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms

Comments

Result of this Calibration Step:
N/A

Date
Jul. 21, 2025

Technician
EMh

Velocity Decoder Calibration Protocol

Part 2: Mechanical End Test

Polytec

Certificate No.: C131563 / L1030877

Device Type: VFX-F-110 **Serial No.:** 0331036

Decoder Type: VFX-V (X)

Reference Conditions:

Frequency[1]**:**	159.17 Hz	**Measurement Procedure:** The laser beam of the vibrometer is directed to the reference surface of a vibration calibrator, which is traceable to national standards for the realization of the physical units. The output of the respective decoder is compared to the reference vibration amplitude.
Amplitude[1]**:**	10.24 mm/s	
Target Temp.:	(25 ± 3)°C	
Actual Temp:	23.1 °C	
Uncertainty[2]**:**	3.00%	

[1]In some measurement ranges, deviating values are possible (refer to table)
[2]The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Keithley 2001	Type:	Metra VC-20
Serial Number:	0661057	Serial Number:	220550
Certificate No.:	3287284	Certificate No.:	52209
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
20.0 mm/s	10.24 mm/s @ 159.17 Hz	1.0240 Vrms	1.0020 Vrms
50.0 mm/s	10.24 mm/s @ 159.17 Hz	0.4096 Vrms	0.4003 Vrms
100 mm/s	10.24 mm/s @ 159.17 Hz	0.2048 Vrms	0.2005 Vrms
200 mm/s	10.24 mm/s @ 159.17 Hz	0.1024 Vrms	0.1002 Vrms
500 mm/s	10.24 mm/s @ 159.17 Hz	0.0410 Vrms	0.0401 Vrms

Comments

Result of this Calibration Step:
No adjustments to the decoder were necessary

Date
Jul. 21, 2025

Technician
EMh

Displacement Decoder Calibration Protocol

Part 1: Electrical Calibration

Polytec

Certificate No.: C131563 / L1030877

Device Type: VFX-F-110

Serial No.: 0331036

Decoder Type: VFX-D (X)

Reference Conditions:

Frequency[1]:	1 kHz	**Measurement Procedure:** Synthetic Doppler signals substitute those which would come from the vibrometer optical system when acquiring a sinusoidal vibration of known frequency and amplitude. The output of the respective decoder is compared to the represented reference vibration amplitude.
Amplitude[1]:	70%	
Target Temp.:	(25 ± 3)°C	
Actual Temp:	23.0 °C	
Uncertainty[2]:	1.00%	

[1]In some measurement ranges, deviating values are possible (refer to table)

[2]The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Tektronix AFG 3252	Type:	Keithley 2001
Serial Number:	C022189	Serial Number:	0661057
Certificate No.:	3292055	Certificate No.:	3287284
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
500 nm	481.25 nm @ 1 kHz	1.3612 Vrms	1.3624 Vrms
1.00 µm	700.00 nm @ 1 kHz	0.9900 Vrms	0.9911 Vrms
2.00 µm	1.4000 µm @ 1 kHz	0.9900 Vrms	0.9908 Vrms
5.00 µm	3.5000 µm @ 1 kHz	0.9900 Vrms	0.9905 Vrms
10.0 µm	7.0000 µm @ 1 kHz	0.9900 Vrms	0.9905 Vrms
20.0 µm	14.000 µm @ 1 kHz	0.9900 Vrms	0.9907 Vrms
50.0 µm	35.000 µm @ 1 kHz	0.9900 Vrms	0.9906 Vrms
100 µm	70.000 µm @ 1 kHz	0.9900 Vrms	0.9906 Vrms
200 µm	140.00 µm @ 1 kHz	0.9900 Vrms	0.9906 Vrms
500 µm	350.00 µm @ 1 kHz	0.9900 Vrms	0.9907 Vrms
1.00 mm	700.00 µm @ 1 kHz	0.9900 Vrms	0.9907 Vrms
2.00 mm	1.4000 mm @ 500 Hz	0.9900 Vrms	0.9912 Vrms
5.00 mm	3.5000 mm @ 250 Hz	0.9900 Vrms	0.9913 Vrms
10.0 mm	7.0000 mm @ 125 Hz	0.9900 Vrms	0.9911 Vrms
20.0 mm	14.000 mm @ 62 Hz	0.9900 Vrms	0.9909 Vrms
50.0 mm	35.000 mm @ 15 Hz	0.9900 Vrms	0.9908 Vrms
100 mm	70.000 mm @ 10 Hz	0.9900 Vrms	0.9908 Vrms
200 mm	70.000 mm @ 10 Hz	0.4950 Vrms	0.4952 Vrms

Comments

Result of this Calibration Step:
No adjustments to the decoder were necessary

Date
Jul. 21, 2025

Technician
EMh

Displacement Decoder Calibration Protocol

Part 2: Mechanical End Test

Certificate No.: C131563 / L1030877

Device Type:	VFX-F-110	**Serial No.:**	0331036
Decoder Type:	VFX-D (X)		

Reference Conditions:

Frequency[1]:	159.17 Hz	**Measurement Procedure:** The laser beam of the vibrometer is directed to the reference surface of a vibration calibrator, which is traceable to national standards for the realization of the physical units. The output of the respective decoder is compared to the reference vibration amplitude.
Amplitude[1]:	10.24 µm	
Target Temp.:	(25 ± 3)°C	
Actual Temp:	22.3 °C	
Uncertainty[2]:	3.00%	

[1]In some measurement ranges, deviating values are possible (refer to table)
[2]The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Keithley 2001	Type:	Metra VC-20
Serial Number:	0661057	Serial Number:	220550
Certificate No.:	3287284	Certificate No.:	52209
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
20.0 µm	10.24 µm @ 159.17 Hz	1.0240 Vrms	1.0028 Vrms
50.0 µm	10.24 µm @ 159.17 Hz	0.4096 Vrms	0.4010 Vrms
100 µm	10.24 µm @ 159.17 Hz	0.2048 Vrms	0.2004 Vrms
200 µm	10.24 µm @ 159.17 Hz	0.1024 Vrms	0.1003 Vrms
500 µm	10.24 µm @ 159.17 Hz	0.0410 Vrms	0.0400 Vrms

Comments

Result of this Calibration Step:
No adjustments to the decoder were necessary

Date	Technician
Jul. 21, 2025	EMh

Alignment of Head Protocol

Polytec

Certificate No.: C131563 / L1030877

Device Type: VFX-I-160 **Serial No.:** 0331024

Optics Check: A separate test is done to check interferometer performance. Various check points are tested to assure the interferometer meets manufacturer specifications. The optical sensitivity is optimized by aligning the interferometer and maximizing signal beam power. The output beam(s) are then checked on a test surface for maximum signal level and a clean, stable signal.

Reference Instruments

Type:	Thor Labs PM100A	Type:	Thor Labs 132C
Serial Number:	P1001255	Serial Number:	220920316
Certificate No.:	491451882025228	Certificate No.:	491451882025228
Due Date:	Feb. 28, 2026	Due Date:	Feb. 28, 2026

Laser Tube:	
1) Output power of beam	9.00 mW
2) Beam is centered in aperture of lens	Pass
3) Beam has correct continuous profile	Pass
Optical Alignment:	
4) Pilot laser and IR laser superimposed at 3+ meters	Pass
5) Pilot laser and IR laser superimposed at short range	Pass
6) Alignment of detector modules	Pass
End Test:	
7) Background signal level without surface reflections	Pass
8) Velocity signal is free of noise and instabilities	Pass

☑ **PASSED VERIFIED**

☐ **FAILED VERIFIED**

Comments

Date Jul. 21, 2025

Technician EMh

~ *Certificate of Calibration and Compliance* ~

Model :	378C10	**Manufacturer :**	PCB
Serial :	178033	**Description :**	1/4" Pressure Microphone and Preamplifier

Microphone Model :	377C10	**Serial :**	370168	**Manufacturer :**	PCB
Preamplifier Model :	426B03	**Serial :**	083097	**Manufacturer :**	PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	5352	CA1247	12/27/2024	12/26/2025
Larson Davis	PRM916	0144	CA2001	12/27/2024	12/26/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	149	CA2072	05/27/2025	05/27/2026
Larson Davis	GPRM902	4162	CA1088	03/12/2025	03/12/2026
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	09/02/2025	09/02/2026
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik

Date: 12/16/2025

ilac-MRA

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CALIBRATION CERT#: 1862.01

PCB PIEZOTRONICS
AN AMPHENOL COMPANY
3425 WALDEN AVENUE - DEPEW, NY 14043
TEL: +1 (888) 684-0013 - FAX: +1 (716) 685-3886 - www.pcb.com

ID: CAL112-3848744283.274+0

~ *Calibration Report* ~

Model : 378C10 **Manufacturer :** PCB

Serial : 178033 **Description :** 1/4" Pressure Microphone and Preamplifier

Microphone Model :	377C10	**Serial :**	370168	**Manufacturer :**	PCB
Preamplifier Model :	426B03	**Serial :**	083097	**Manufacturer :**	PCB

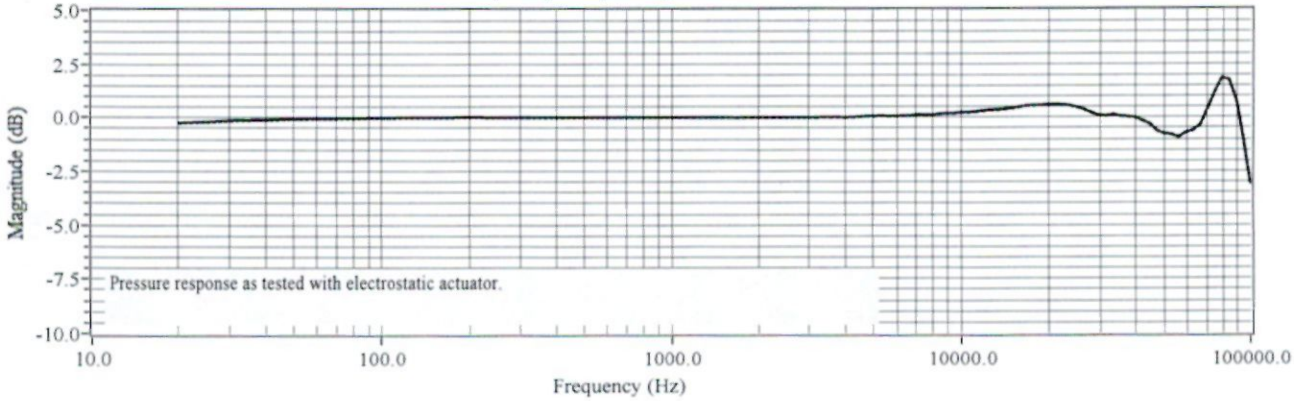
Calibration Data

System Sensitivity at 251.2 Hz : 0.98 mV/Pa
-60.17 dB re 1 V/Pa

Polarization Voltage, External : 0 V

Temperature: 68 °F (20 °C) Ambient Pressure: 994 mbar Relative Humidity: 30 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.24	1584.90	-0.00	6683.40	0.09	28250.00	0.20
25.10	-0.20	1678.80	-0.02	7079.50	0.13	29750.00	0.10
31.60	-0.12	1778.30	-0.00	7498.90	0.11	31500.00	0.08
39.80	-0.11	1883.60	-0.00	7943.30	0.12	33500.00	0.14
50.10	-0.06	1995.30	0.01	8414.00	0.16	35500.00	0.06
63.10	-0.05	2113.50	-0.00	8912.50	0.18	37500.00	0.04
79.40	-0.03	2238.70	-0.00	9440.60	0.20	39750.00	-0.01
100.00	-0.02	2371.40	-0.00	10000.00	0.23	42250.00	-0.17
125.90	-0.00	2511.90	0.00	10592.50	0.24	44750.00	-0.31
158.50	-0.00	2660.70	0.00	11220.20	0.28	47250.00	-0.62
199.50	0.01	2818.40	0.00	11885.00	0.32	50000.00	-0.75
251.20	0.00	2985.40	0.00	12589.30	0.34	53000.00	-0.79
316.20	0.01	3162.30	-0.01	13335.20	0.35	56250.00	-0.92
398.10	-0.00	3349.70	0.01	14125.40	0.39	59500.00	-0.70
501.20	0.01	3548.10	0.01	14962.40	0.43	63000.00	-0.60
631.00	-0.00	3758.40	0.02	15848.90	0.49	66750.00	-0.36
794.30	0.01	3981.10	0.00	16788.00	0.54	70750.00	0.41
1000.00	-0.00	4217.00	0.01	17782.80	0.56	75000.00	1.21
1059.30	0.00	4466.80	0.04	18836.50	0.58	79500.00	1.83
1122.00	0.01	4731.50	0.06	19952.60	0.61	84250.00	1.74
1188.50	-0.01	5011.90	0.05	21250.00	0.60	89250.00	0.79
1258.90	-0.01	5308.80	0.07	22500.00	0.59	94500.00	-1.02
1333.50	-0.01	5623.40	0.06	23750.00	0.54	100000.00	-3.11
1412.50	-0.00	5956.60	0.05	25000.00	0.48		
1496.20	-0.00	6309.60	0.07	26500.00	0.37		

Technician: Leonard Lukasik Date: 12/16/2025

ilac-MRA

A2LA ACCREDITED CALIBRATION CERT# 1862.01

PCB PIEZOTRONICS
AN AMPHENOL COMPANY
3425 WALDEN AVENUE - DEPEW, NY 14043
TEL: +1 (888) 684-0013 - FAX: +1 (716) 685-3886 - www.pcb.com

ID: CAL112-3848744283.274+0

~ *Certificate of Calibration and Compliance* ~

Model :	378C10	**Manufacturer :**	PCB
Serial :	178034	**Description :**	1/4" Pressure Microphone and Preamplifier

Microphone Model :	377C10	**Serial :**	369923	**Manufacturer :**	PCB
Preamplifier Model :	426B03	**Serial :**	083112	**Manufacturer :**	PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	10/17/2024	04/17/2026
Larson Davis	PRM915	0131	CA1205	05/19/2025	05/19/2026
Larson Davis	PRM902	5352	CA1247	12/27/2024	12/26/2025
Larson Davis	PRM916	0144	CA2001	12/27/2024	12/26/2025
Larson Davis	CAL250	5569	CA2284	12/23/2024	12/23/2025
Larson Davis	2201	149	CA2072	05/27/2025	05/27/2026
Larson Davis	GPRM902	4162	CA1088	03/12/2025	03/12/2026
Larson Davis	PRM915	123	CA866	01/07/2025	01/07/2026
Larson Davis	PRA951-4	0241	CA1449	09/02/2025	09/02/2026
Bruel & Kjaer	4192	2764626	CA1636	03/25/2025	03/25/2026
Newport	iTHX-SD/N	1080002	CA1511	02/07/2025	02/07/2026
PCB	68510-02	N/A	CA2672	02/06/2025	02/06/2026

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.25 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik   Date: 12/16/2025

ilac-MRA ACCREDITED CALIBRATION CERT# 1862.01

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ID: CAL112-3848747117.949+0

~ *Calibration Report* ~

Model : 378C10 **Manufacturer :** PCB

Serial : 178034 **Description :** 1/4" Pressure Microphone and Preamplifier

Microphone Model : 377C10 **Serial :** 369923 **Manufacturer :** PCB

Preamplifier Model : 426B03 **Serial :** 083112 **Manufacturer :** PCB

Calibration Data

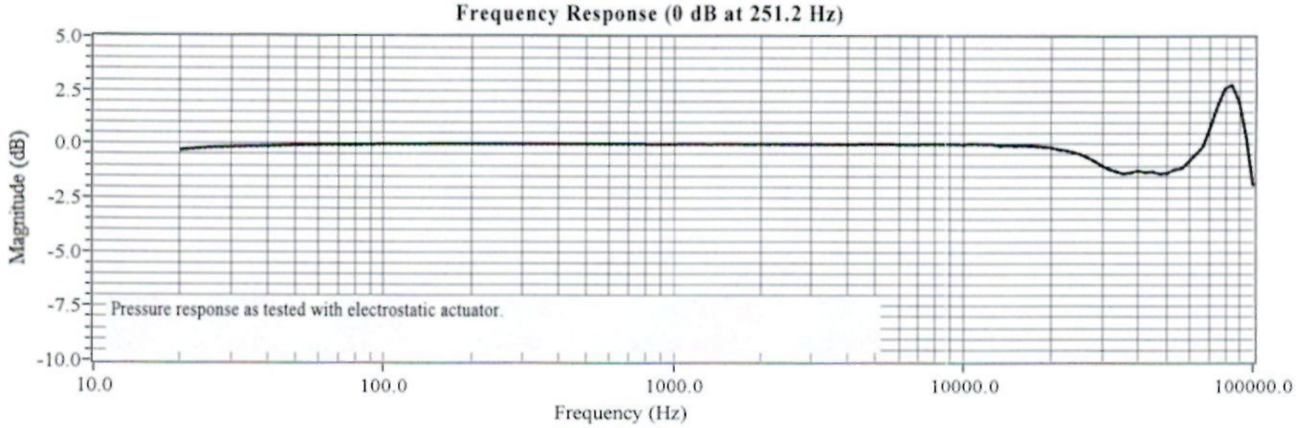
System Sensitivity at 251.2 Hz : 0.97 mV/Pa

-60.26 dB re 1 V/Pa

Polarization Voltage, External : 0 V

Temperature: 68 °F (20 °C) Ambient Pressure: 994 mbar Relative Humidity: 30 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)	Frequency (Hz)	Pressure (dB)
20.00	-0.30	1584.90	-0.01	6683.40	-0.02	28250.00	-0.76
25.10	-0.19	1678.80	0.00	7079.50	-0.01	29750.00	-0.95
31.60	-0.15	1778.30	-0.01	7498.90	-0.01	31500.00	-1.13
39.80	-0.11	1883.60	-0.02	7943.30	-0.00	33500.00	-1.26
50.10	-0.07	1995.30	0.00	8414.00	-0.00	35500.00	-1.36
63.10	-0.04	2113.50	-0.01	8912.50	0.00	37500.00	-1.31
79.40	-0.04	2238.70	-0.01	9440.60	-0.01	39750.00	-1.22
100.00	-0.02	2371.40	-0.02	10000.00	-0.02	42250.00	-1.30
125.90	-0.01	2511.90	-0.01	10592.50	0.01	44750.00	-1.27
158.50	-0.00	2660.70	-0.01	11220.20	0.00	47250.00	-1.36
199.50	-0.00	2818.40	-0.02	11885.00	-0.00	50000.00	-1.35
251.20	0.00	2985.40	-0.01	12589.30	-0.01	53000.00	-1.18
316.20	0.00	3162.30	-0.01	13335.20	-0.08	56250.00	-1.13
398.10	0.00	3349.70	-0.02	14125.40	-0.04	59500.00	-0.81
501.20	-0.00	3548.10	-0.02	14962.40	-0.07	63000.00	-0.47
631.00	0.00	3758.40	-0.02	15848.90	-0.06	66750.00	-0.11
794.30	-0.00	3981.10	-0.02	16788.00	-0.07	70750.00	0.80
1000.00	-0.01	4217.00	-0.02	17782.80	-0.09	75000.00	1.81
1059.30	0.00	4466.80	-0.00	18836.50	-0.11	79500.00	2.60
1122.00	0.00	4731.50	-0.01	19952.60	-0.15	84250.00	2.78
1188.50	-0.00	5011.90	0.00	21250.00	-0.23	89250.00	1.98
1258.90	-0.01	5308.80	-0.00	22500.00	-0.28	94500.00	0.34
1333.50	-0.01	5623.40	0.00	23750.00	-0.36	100000.00	-1.91
1412.50	-0.00	5956.60	-0.02	25000.00	-0.44		
1496.20	-0.00	6309.60	-0.02	26500.00	-0.58		

Technician: Leonard Lukasik Date: 12/16/2025

ilac-MRA ACCREDITED CALIBRATION CERT#: 1862.01

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ID: CAL112-3848747117.949+0

~ *Certificate of Calibration and Compliance* ~

Model :	378B02	**Manufacturer :**	PCB
Serial :	179741	**Description :**	1/2" Free-Field Microphone and Preamplifier

Microphone Model :	377B02	**Serial :**	372797	**Manufacturer :**	PCB
Preamplifier Model :	426E01	**Serial :**	096215	**Manufacturer :**	PCB

Calibration Environmental Conditions

Environmental test conditions as printed on microphone calibration chart.

Reference Equipment

Manufacturer	Model #	Serial #	Control #	Cal Date	Due Date
National Instruments	PCIe-6351	01896F08	CA1918	04/17/2026	10/15/2027
Larson Davis	PRM915	0150	CA2116	01/26/2026	01/26/2027
Larson Davis	PRM902	5300	CA1627	11/04/2025	11/04/2026
Larson Davis	PRM916	140	CA2012	11/03/2025	11/03/2026
Larson Davis	CAL250	5026	CA1278	09/04/2025	09/04/2026
Larson Davis	2201	150	CA2228	12/02/2025	12/02/2026
Larson Davis	GPRM902	4923	CA2237	12/01/2025	12/01/2026
Larson Davis	PRM915	147	CA2179	12/02/2025	12/02/2026
Larson Davis	PRA951-4	0241	CA1449	09/02/2025	09/02/2026
Bruel & Kjaer	4192	3259548	CA3533	01/29/2026	01/29/2027
Newport	iTHX-SD/N	1080002	CA1511	03/31/2026	03/31/2027
		N/A	N/A	N/A	N/A

Frequency sweep performed with B&K UA0033 electrostatic actuator.

Condition of Unit

As Found : n/a

As Left : New Unit, In Tolerance

Notes

1. Calibration of reference equipment is traceable to one or more of the following National Labs; NIST, PTB or DFM.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. System Sensitivity is measured following procedure AT603-5.
6. Measurement uncertainty (95% confidence level with coverage factor of 2) for sensitivity is +/-0.20 dB.
7. Unit calibrated per ACS-63.
8. Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Technician: Leonard Lukasik   Date: 05/08/2026

ilac-MRA

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ID: CAL112-3861107745.584+0

~ *Calibration Report* ~

Model : 378B02 **Manufacturer :** PCB

Serial : 179741 **Description :** 1/2" Free-Field Microphone and Preamplifier

Microphone Model : 377B02 **Serial :** 372797 **Manufacturer :** PCB

Preamplifier Model : 426E01 **Serial :** 096215 **Manufacturer :** PCB

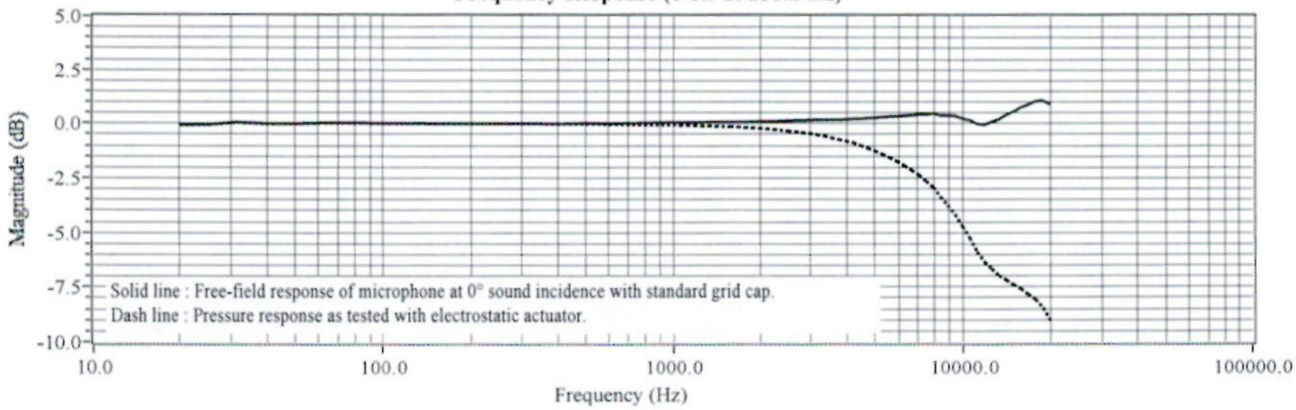
Calibration Data

System Sensitivity at 251.2 Hz : 49.15 mV/Pa
-26.17 dB re 1 V/Pa

Polarization Voltage, External : 0 V

Temperature: 68 °F (20 °C) Ambient Pressure: 985 mbar Relative Humidity: 36 %

Frequency Response (0 dB at 251.2 Hz)



Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)	Frequency (Hz)	Pressure (dB)	Free-Field (dB)
20.00	-0.07	-0.07	1584.90	-0.14	0.07	6683.40	-2.14	0.38
25.10	-0.04	-0.04	1678.80	-0.15	0.08	7079.50	-2.36	0.42
31.60	0.06	0.06	1778.30	-0.18	0.07	7498.90	-2.66	0.41
39.80	0.00	0.00	1883.60	-0.19	0.09	7943.30	-2.93	0.46
50.10	-0.00	-0.00	1995.30	-0.21	0.10	8414.00	-3.36	0.37
63.10	0.02	0.02	2113.50	-0.24	0.10	8912.50	-3.74	0.37
79.40	0.03	0.03	2238.70	-0.27	0.10	9440.60	-4.17	0.35
100.00	0.01	0.01	2371.40	-0.30	0.11	10000.00	-4.72	0.23
125.90	0.01	0.01	2511.90	-0.33	0.13	10592.50	-5.24	0.16
158.50	-0.00	-0.00	2660.70	-0.36	0.15	11220.20	-5.88	-0.02
199.50	0.00	0.00	2818.40	-0.41	0.15	11885.00	-6.37	-0.05
251.20	0.00	0.00	2985.40	-0.45	0.17	12589.30	-6.67	0.10
316.20	-0.01	0.00	3162.30	-0.50	0.18	13335.20	-6.99	0.20
398.10	-0.02	-0.02	3349.70	-0.56	0.18	14125.40	-7.19	0.40
501.20	-0.02	0.02	3548.10	-0.65	0.17	14962.40	-7.40	0.57
631.00	-0.03	0.01	3758.40	-0.72	0.18	15848.90	-7.59	0.76
794.30	-0.05	0.04	3981.10	-0.81	0.19	16788.00	-7.84	0.88
1000.00	-0.07	0.05	4217.00	-0.88	0.23	17782.80	-8.08	1.03
1059.30	-0.08	0.05	4466.80	-1.01	0.22	18836.50	-8.47	1.04
1122.00	-0.08	0.06	4731.50	-1.12	0.25	19952.60	-9.02	0.91
1188.50	-0.09	0.06	5011.90	-1.26	0.27			
1258.90	-0.10	0.06	5308.80	-1.40	0.30			
1333.50	-0.10	0.08	5623.40	-1.56	0.32			
1412.50	-0.12	0.07	5956.60	-1.74	0.33			
1496.20	-0.13	0.07	6309.60	-1.94	0.35			

Technician: Leonard Lukasik Date: 05/08/2026

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ilac-MRA ACCREDITED CALIBRATION CERT#: 1862.01

ID: CAL112-3861107745.584+0

~ Calibration Certificate ~

Model Number: 086E80

Serial Number: 54385

Description: Impulse Force Hammer

Manufacturer: PCB

Customer:

P.O. Number:

Method: Impulse (AT303-1)

Calibration Data

Output Bias: 10.2 VDC

Temperature: 71 °F = 22 °C

Relative Humidity: 41 %

Hammer Configuration		Hammer Sensitivity*	
Tip	**Extender**	**mV/lb**	**mV/N**
Steel	None	99.96	22.47
Vinyl	Steel	84.62	19.03
Vinyl	None	80.87	18.18

*Above data is valid for all supplied tips.

Condition of Unit

As Found: N/A

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater.

Notes:

1. Calibration is N.I.S.T. Traceable through Project 684/O-0000069046 and PTB Traceable through Project 17036.
2. This certificate may not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI/NCSL Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with a coverage factor of 2) is +/-3.8%.

Technician: Richard Gardner RG

Date: 5/11/2026

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CAL13-3861350067.419+0

~ Calibration Certificate ~

Per ISO 16063-21

Model Number: **356A03**

Serial Number: **LW458895 (x axis)**

Description: ICP® Triaxial Accelerometer

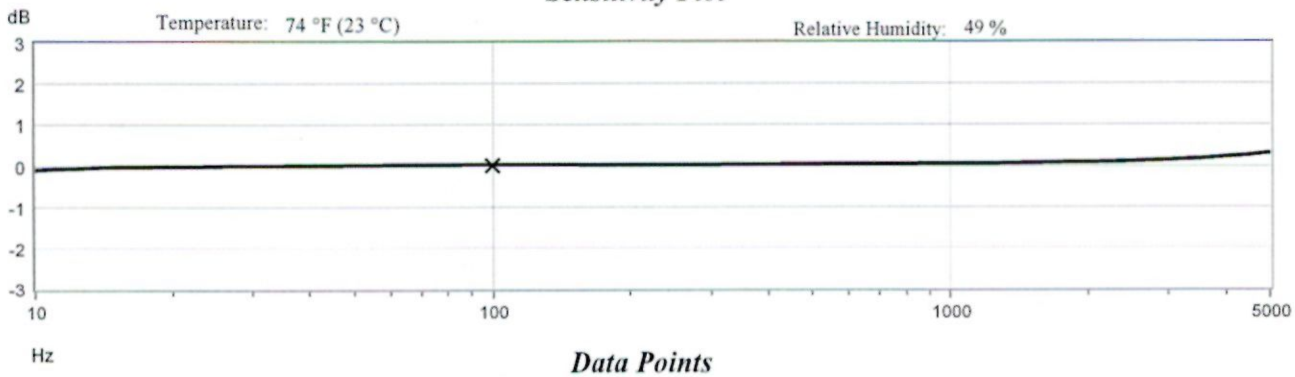
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz	**10.34 mV/g**	Output Bias	11.7 VDC
	(1.055 mV/m/s²)	Transverse Sensitivity	2.0 %
Discharge Time Constant	0.40 seconds		

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-1.2	300	0.2
15	-0.4	500	0.4
30	-0.1	1000	0.6
50	0.1	3000	1.5
REF. FREQ.	0.0	5000	3.6

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Inverted Vertical

Acceleration Level (pk)[1]: 10.0 g (98.1 m/s²)

[1]The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².

[2]The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

1. Calibration is NIST Traceable thru Project 684/O-0000069046 and PTB Traceable thru Project 17036.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows: 5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

SLE 6124

Technician: Simeon Eason Date: 5/21/2026

ilac-MRA

ACCREDITED CALIBRATION CERT#: 1862.02

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CAL48-3862250197.979+0

ACS-1T

~ Calibration Certificate ~

Per ISO 16063-21

Model Number: **356A03**

Serial Number: **LW458895 (y axis)**

Description: ICP® Triaxial Accelerometer

Manufacturer: PCB

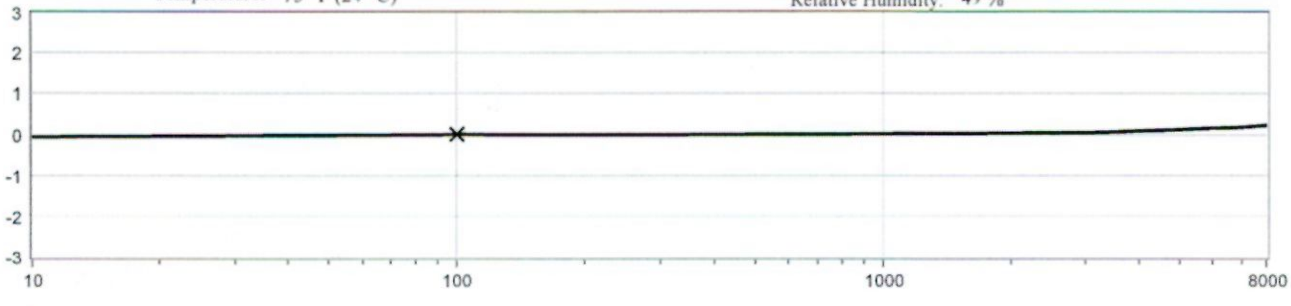
Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz	**10.25 mV/g**	Output Bias	11.4 VDC
	(1.045 mV/m/s²)	Transverse Sensitivity	0.4 %
Discharge Time Constant	0.37 seconds		

Sensitivity Plot

Temperature: 75 °F (24 °C) Relative Humidity: 49 %



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-0.8	300	-0.2	7000	2.0
15	-0.6	500	0.0	8000	2.5
30	-0.4	1000	0.1		
50	-0.3	3000	0.5		
REF. FREQ.	0.0	5000	1.3		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical

Acceleration Level (pk)[1]: 10.0 g (98.1 m/s²)

[1]The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x (freq)².

[2]The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s².

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

1. Calibration is NIST Traceable thru Project 684/O-0000069046 and PTB Traceable thru Project 17036.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows: 5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician: Simeon Eason SLE 6124 Date: 5/21/2026

ilac-MRA ACCREDITED CALIBRATION CERT#: 1862.02

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CAL48-3862248076.178+0

ACS-1T

~ Calibration Certificate ~

Per ISO 16063-21

Model Number: 356A03

Serial Number: LW458895 (z axis)

Description: ICP® Triaxial Accelerometer

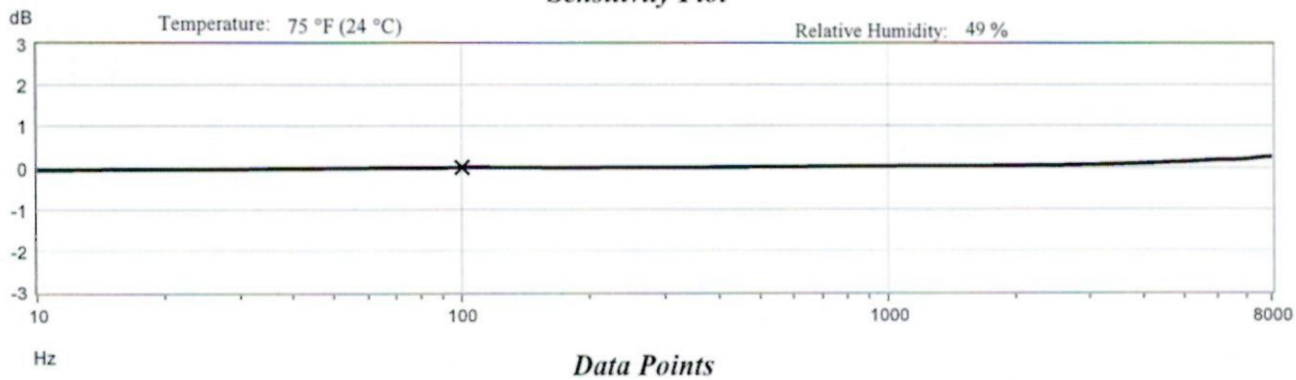
Manufacturer: PCB

Method: Back-to-Back Comparison AT401-3

Calibration Data

Sensitivity @ 100 Hz	**10.94 mV/g**	Output Bias	11.5 VDC
	(1.116 mV/m/s²)	Transverse Sensitivity	1.1 %
Discharge Time Constant	0.41 seconds		

Sensitivity Plot



Data Points

Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)	Frequency (Hz)	Dev. (%)
10	-0.7	300	-0.1	7000	2.2
15	-0.6	500	0.1	8000	2.9
30	-0.5	1000	0.2		
50	-0.3	3000	0.7		
REF. FREQ.	0.0	5000	1.5		

Mounting Surface: Tungsten Adapter Fastener: Adhesive Fixture Orientation: Vertical

Acceleration Level (pk)[1]: 10.0 g (98.1 m/s^2)

[1]The acceleration level may be limited by shaker displacement at low frequencies. If the listed level cannot be obtained, the following formula is used to set the vibration amplitude: Acceleration Level (g) = 0.00785 x $(freq)^2$.

[2]The gravitational constant used for calculations by the calibration system is: 1 g = 9.80665 m/s^2.

Condition of Unit

As Found: n/a

As Left: New Unit, In Tolerance

Where provided, statements of conformity are made in accordance with Simple Acceptance decision rule as defined in ILAC G8 with TUR of 4:1 or greater

Notes

1. Calibration is NIST Traceable thru Project 684/O-0000069046 and PTB Traceable thru Project 17036.
2. This certificate shall not be reproduced, except in full, without written approval from PCB Piezotronics, Inc.
3. Calibration is performed in compliance with ISO 10012-1, ANSI Z540.3 and ISO 17025.
4. Measurement results relate only to the items tested. Refer to Manufacturer's Specification Sheet for performance details.
5. Measurement uncertainty (95% confidence level with coverage factor of 2) for frequency ranges tested during calibration are as follows: 5-9 Hz; +/- 2.0%, 10-99 Hz; +/- 1.5%, 100-1999 Hz; +/- 1.0%, 2-10 kHz; +/- 2.5%, 10-15 kHz; +/- 5%.

Technician: Simeon Eason SLE 6124 Date: 5/21/2026

ilac-MRA ACCREDITED CALIBRATION CERT#: 1862.02

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AN AMPHENOL COMPANY

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CAL48-3862248271.303+0

ACS-1T